Research Gaps and Proposed Approach: Temporal Attack Graphs for Post-Incident Network Forensics

Executive Overview

Attack graphs have traditionally been used for **proactive vulnerability management** — identifying host configurations to patch or firewalls to reconfigure before an attack occurs. Our previous work (*Gain & Barik, COSE 2026*) introduced **Temporal Attack Graphs (TAGs)** $(\mathcal{G}^a(t_{min}, t_{max}) = (G_0, G_1, \dots, G_{\tau-1}))$ to model time-varying vulnerability surfaces and quantify dynamic security risks via temporal path distance ((d^a_{ij})) and centralities ($(\mathcal{B}^a_i, \mathcal{C}^a_i)$).

This research shifts the paradigm from **proactive hardening** to **reactive post-incident forensic reconstruction**. When a network intrusion occurs and a compromise is confirmed at a target host (j^*), *forensic investigators face critical challenges:*

1. What was the exact path the attacker took?
2. How far back in time must we search the log files?
3. Which unverified potential exploits lack supporting forensic evidence?
4. How confident can we be in our reconstructed attack scenario under courtroom cross-examination?*

Existing forensic literature (Wang & Daniels 2008, Liu et al. 2012, Zhang et al. 2015, He et al. 2016, Barrère et al. 2017) fails to address these challenges on dynamic networks. This document details the **5 critical research gaps** identified across current literature and presents our **unified theoretical approach** to solve them.

Part 1: Comprehensive Gap Analysis

The table below summarizes the 5 key research gaps, comparing prior state-of-the-art approaches against our proposed solutions:

#	Research Gap	State-of-the-Art Limitations	Baseline Papers	Our Proposed Approach
1	**No Temporal Centrality Trajectories for Dynamic Triage**	Centralities are computed once over static or fully-aggregated graphs (*(Ax = \lambda x)*, *(y=H^T Z)*). They cannot capture how host criticality shifts across time windows during an active investigation.	Wang & Daniels (2008), He et al. (2016), Gain & Barik (2026)	**Windowed Centrality Trajectories** *(\vec{\mathcal{B}}^a_i, \vec{\mathcal{C}}^a_i)*: Evaluates host centrality over window prefixes *(W_1 \subset W_2 \subset \dots \subset W_k)* to detect newly emerging relay points.
2	**No Time-Indexed Evidence Verification & Gap Metric**	Relevancy *(r(e))* is static at edge creation. Models do not track incremental evidence arrival or quantify "missing evidence detours."	Wang & Daniels (2008) (*(r(e) \in {0, 0.5, 1})*), COSE 2026 (No evidence layer)	**Time-Indexed Evidence Relevancy** *(r^a(e, W_p))*, **Forensic Subgraph** *(\mathcal{G}^{a,\text{forensic}}(W_p))*, and **Evidence Gap Metric** *(\Delta^a_{ij}(W_p) = d^{a,\text{forensic}}_{ij} - d^a_{ij})*.
3	**No Ranked Backward Temporal Path Reconstruction**	Existing methods use forward-only traversal, return scalar bounds, or extract a single greedy static path without temporal window ordering (*(W_b \le W_p)*).	Zhang et al. (2015) (`AdjustSuggestion`), Barrère et al. (2017) (`traceBack`), COSE 2026 (Forward BFS)	**Backward Temporal Path Set** *(Q^{a, \text{back}}_{j^})* **& Algorithm 1** *(`BackwardReconstruct`)*: Reverses window constraints to produce a ranked set of plausible backward paths using score* *(\pi(p) = \alpha \pi_{\text{struct}} + (1-\alpha) \pi_{\text{evid}})*.
4	**No Principled Search-Scope Bound**	Forensic search windows are selected arbitrarily or via	Wang & Daniels (2008) (Self-extending window *(T)*)	**Lookback Bound Theorem** (*(\lambda^_{j^} \le t^ - t^{\text{first}}_{j^})*): Derives a provably sufficient historical search window from TAG

#	Research Gap	State-of-the-Art Limitations	Baseline Papers	Our Proposed Approach
	(Lookback Window)	fixed ad-hoc time limits (e.g., $T=60\text{s}$), risking either missing early attack steps or over-searching logs.		reachability and Capability Persistence Axiom.
5	**No Forensic Confidence & Evidentiary Defensibility Metrics**	Forensic reconstructions present a single route without quantifying uncertainty or identifying single-point-of-failure evidence links under evidentiary challenge.	All 5 prior papers (Zero formal confidence metrics)	**Temporal AG Entropy** (H^a_{j}), *Entropy Reduction* (ΔH^a_{j}), and **Reconstruction Robustness** ($\rho^a_{j}(e)$) *via Algorithm 2* (`ComputeForensicConfidence`)*.

Part 2: Detailed Technical Gaps & Baseline Literature Limitations

Gap 1: Static Centrality vs. Dynamic Forensic Triage

- **The Problem**: In dynamic enterprise networks, firewall rules change and new vulnerabilities open up over time. A host that was harmless at window w_0 may become a critical pivot host at window w_2.
- **Prior Work Deficiencies**:
- *Wang & Daniels (2008)* compute static Eigenvector Centrality $Ax = \lambda x$ over the full evidence graph.
- *He et al. (2016)* iterate Nodeworth/Linkworth matrix updates to convergence on a static graph.
- *Gain & Barik (2026)* compute TAG betweenness $\mathcal{B}^a_i$ once over the entire time span $[t_{min}, t_{max}]$.
- **Impact**: Investigators cannot see *when* a host became critical, leading to inefficient log review prioritization.

Gap 2: Potential Exploits vs. Verified Evidence

- **The Problem**: An attack graph represents *potential* exploit paths based on network connectivity and software vulnerabilities. However, in post-incident forensics, an investigator only cares about paths supported by actual log files, IDS alerts, or disk artifacts.
- **Prior Work Deficiencies**:
- *Wang & Daniels (2008)* assign static ternary relevancy $(r(e) \in \{0, 0.5, 1\})$ at edge creation. As new logs arrive at later windows, relevancy cannot evolve.
- *COSE 2026 base model* contains no evidence layer — it treats all topological edges as equally real.
- **Impact**: No formal way to measure the "evidence gap" ((Δ^a_{ij})) between what *could* happen topologically versus what is *proven* to have happened.

Gap 3: Forward Reachability vs. Backward Incident Tracing

- **The Problem**: Post-incident forensics starts at the **end** of an attack — a compromised host $(j^)$ *discovered at window* $(t^)$. Forensic analysts must trace **backward** in time to find the entry point (patient zero) and lateral steps.
- **Prior Work Deficiencies**:
- *COSE 2026 Algorithm 1* computes *forward* shortest paths from an initial source host (i).
- *Zhang et al. (2015) Algorithm 3* performs backward BFS but outputs only two scalars (`min-dst-time` and `max-src-time`) to fix timestamp falsification — it does not reconstruct attack paths.
- *Barrère et al. (2017) Algorithm 2 (* `traceBack` *)* traces backward but operates on a static (non-temporal) graph, returning a single path based on greedy local choices.
- **Impact**: Lack of a temporal backward reconstruction algorithm that yields a ranked candidate set of non-decreasing time paths ($(W_b \le W_p)$).

Gap 4: Arbitrary Time Bounds vs. Graph-Derived Lookback Bounds

- **The Problem**: How far back in time should a forensic analyst query SIEM/log systems? Searching 30 days of logs across 10,000 hosts is computationally prohibitive; searching 2 hours might miss the attacker's initial breach weeks prior.
- **Prior Work Deficiencies**:
- *Wang & Daniels (2008) Algorithm 1* relies on a manually configured hyperparameter (T) (e.g., $(T = 60\text{s})$) to group alerts.
- No existing paper provides a mathematically proven bound for forensic lookback search.
- **Impact**: Forensic scope selection remains an ad-hoc guessing game.

Gap 5: Absence of Forensic Confidence & Courtroom Defensibility Quantification

- **The Problem**: In legal proceedings or executive briefings, an investigator cannot simply state "we believe path X occurred." They must answer: *How certain are you?* and *What happens to your conclusion if one log entry is ruled inadmissible?*

- **Prior Work Deficiencies**: None of the 5 baseline papers quantify reconstruction uncertainty or perform sensitivity analysis on evidence edges.
- **Impact**: Forensic reporting lacks quantitative defensibility under evidentiary challenge.

Part 3: Our Proposed Solution & Formal Framework

Our proposed framework builds directly on the base TAG formalism $(\mathcal{G}^a(t_{min}, t_{max}) = (G_0, \dots, G_{\tau-1}))$ from *Gain & Barik (COSE 2026)* and introduces an end-to-end forensic reconstruction engine:

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│  INPUT: TAG G^a, confirmed target j*, investigation window t*, evidence r^a │
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│                                                                          │
│  Step 1:  Compute Lookback Bound λ*_{j*}                [Theorem 1]      │
│           → Determines provably sufficient historical search window      │
│                                                                          │
│  Step 2:  Construct Forensic Subgraph G^{a,forensic}     [Definition 5]  │
│           → Restricts analysis to edges with confirmed evidence (r^a=1)  │
│                                                                          │
│  Step 3:  Run BackwardReconstruct(G^a, j*, t*, r^a, K)  [Algorithm 1]    │
│           → Generates ranked candidate paths Q^{a,back}_{j*}             │
│                                                                          │
│  Step 4:  Compute Windowed Centrality Trajectories       [Definitions 8-9] │
│           → Tracks B^a_i(W_l) over time prefixes for dynamic triage      │
│                                                                          │
│  Step 5:  Compute Forensic Confidence                    [Algorithm 2]   │
│           → Quantifies Entropy H^a_{j*}, Robustness ρ^a_{j*}(e), Gap Δ^a │
│                                                                          │
│  Step 6:  Characterize Attack Dynamics                  [Definitions 10-11] │
│           → Computes Velocity v^a (automated vs APT) & Persistence Π^a   │
│                                                                          │
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│  OUTPUT: Ranked attack paths, confidence bounds, courtroom-defensible metrics │
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```

Key Theoretical Constructs Implemented in `paper/draft_sections_3_5.md`:

1. Time-Indexed Evidence Layer & Evidence Gap (§3.2)

- **Time-Indexed Evidence Relevancy** $(r^a(e, W_p) \in \{0, 1\})$: Models incremental evidence collection up to investigation window (W_p).
- **Forensic Subgraph** $(\mathcal{G}^{a,\text{forensic}}(W_p))$: Subgraph containing only evidenced edges ($(r^a = 1)$).

- **Evidence Gap Metric** *(\Delta^a_{ij}(W_p) = d^{a,\text{forensic}}{ij}(W_p) - d^a{ij})*: Quantifies missing evidence links forcing detours.

2. Windowed Centrality Trajectories (§3.3)

- **Trajectory** *(\vec{\mathcal{B}}^a_i = (\mathcal{B}^a_i(W_1), \dots, \mathcal{B}^a_i(W_k)))*: Computes betweenness centrality over expanding window prefixes *(W_l = {0, \dots, l-1})* to identify newly emerging chokepoints during active investigations.

3. Backward Temporal Path Reconstruction (§4)

- **Backward Path Set** *(Q^{a,\text{back}}{j^}): Enumerates paths from entry points (V_{\text{ext}}) to target (j^) respecting reverse non-decreasing window constraints ((W_b(t_l) \le W_p(t*{l+1}))* for *(t \le t^*)*).
- **Plausibility Scoring**:

$$\pi(p) = \alpha \cdot \pi_{\text{struct}}(p) + (1-\alpha) \cdot \pi_{\text{evid}}(p)$$

combining windowed intermediate betweenness centrality with evidence fraction.
- **Algorithm 1 (`BackwardReconstruct`)**: Bounded priority-queue traversal running in *(O(K \cdot (n+m) \cdot \tau))* time.

4. The Lookback Bound Theorem (§5.1)

- **Capability Persistence Axiom**: Attacker access persists once gained unless remediated (*(\text{access}(v, t) = 1 \implies \text{access}(v, t') = 1 \ \forall t' \ge t)*).
- **Theorem 1**: Derives the provable minimum lookback window *(\lambda^_{j^})*:

$$\lambda^*_{j^*} \le t^* - t^{\text{first}}_{j^*}$$

where *(t^{\text{first}}_{j^}) is the earliest temporal reachability window of target (j^)*.

5. Forensic Confidence & Defensibility Metrics (§5.2 – §5.3)

- **Temporal AG Entropy** *(H^a_{j^*})*: Shannon entropy over normalized path plausibilities *(\hat{\pi}(p))*. *(H^a = 0)* indicates zero ambiguity; *(H^a = \log_2 |Q|)* indicates maximum uncertainty.
- **Entropy Reduction** *(\Delta H^a_{j^}(W_p) = H^a_{j^} - H^{a,\text{forensic}}_{j^*}(W_p))*: Measures the exact information value (in bits) of collected evidence.
- **Reconstruction Robustness** *(\rho^a_{j^}(e) = 1 - \frac{|Q^{a,\text{back}}_{j^}(e)|}{|Q^{a, \text{back}}{j^ prob}|}): Leave-one-out sensitivity testing. If (\rho^a_{j^}(e) = 0), edge (e) is a single point of failure; if (\rho^a*{j^*}(e) > 0.5)*, the conclusion is robust to losing evidence edge *(e)*.
- **Algorithm 2 (`ComputeForensicConfidence`)**: Computes entropy and robustness profiles in *(O(K \cdot L_{\max}))* time.

Summary & Next Steps

This framework directly bridges the gap between theoretical attack graph modeling and real-world digital forensics:
- It grounds search scope in **graph theory** (Lookback Bound Theorem) rather than arbitrary limits.
- It transforms forensic reconstruction from **opaque guessing** to **probabilistic path ranking** (Plausibility Score & Backward Algorithm).
- It provides **quantifiable courtroom defensibility** (Entropy and Robustness).

Phased Roadmap:

1. ✅ **Formal Theory Draft**: Sections 3–5 completed in `paper/draft_sections_3_5.md`.
2. 🔄 **Implementation**: Code algorithms in Python / Neo4j extending the COSE pipeline.
3. 🔄 **Validation**: Run experiments on multi-window synthetic datasets and DARPA 2000 LLDOS benchmark (Section 6).
4. 🔄 **Submission**: Finalize paper for IFIP WG 11.9 (2027) deadline.